P0AA6-F0、 P0AA6-F1、 P0AA6-F2、 P0AA7-00

Fault diagnosis code
P0AA6-F0：Insulation detection1level failure
P0AA6-F1：Insulation detection2level failure
P0AA6-F2：Insulation detection3level failure
P0AA7-00：Insulation detection circuit hardware failure
Detection tools
1 Multimeter, diagnostic apparatus and tramegger.
Possible causes
• BMS fault • Cooling system fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P0AA6-F0
The insulation value of the high voltage system is lower than the fault threshold: insulation resistance value< 1000Ω/V
P0AA6-F1
The insulation value of the high voltage system is lower than the fault threshold: insulation resistance value< 600Ω/V
P0AA6-F2
The insulation value of the high voltage system is lower than the fault threshold: insulation resistance value< 200Ω/V
P0AA7-00
HVU Hardware failure (abnormal voltage insulation acquisition circuit)
To set the effect of a fault code condition
High voltage component does not work properly

Troubleshooting steps	
Step 1	General inspection.
	1 Check battery capacity. 2 Check the harness plug of BMS for damage, poor contact, aging and looseness. 3 Check the harness plug of battery pack high voltage for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the BMS with diagnostic apparatus. 4 Is there any other fault code except for P0AA6-F0、P0AA6-F1、P0AA6-F2、P0AA7-00?
	YES → Refer to:DTC Summary list(BMS).
NO ↓	
Step 3	Check the connecting cable of BMS.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Disconnect the harness plug battery pack high voltage. 4 Disconnect the high voltage harness plug of BMS. 5 Check whether the BMS connection cable is rotten, damaged, and water enters. 6 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 4	Check the insulation of BMS.
	1 Use a megohm meter to check the insulation resistance of the BMS housing and grounding according to the procedure in the maintenance manual. **Standard value （insulation level 1) ＞ 1000Ω/V** **Standard value （insulation level 2) ＞ 600Ω/V** 2 Check whether the result is normal or not?

	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Check the insulation of BMS high-voltage cables.
	1 Use a megohm meter to check the insulation resistance of the BMS high voltage and grounding according to the procedure in the maintenance manual. **Standard value （insulation level 1) ＞ 1000Ω/V** **Standard value （insulation level 2) ＞ 600Ω/V** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 6	Perform initialization.
	1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 The BMS module was initialized with the diagnostic instrument. 5 Can the BMS be initialized and is the BMS restored?
	NO → Repair or change the BMS module.
YES ↓	
Step 7	Test whether BMS is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug CH1001 of BMS. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BMS plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BMS.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	

Diagnosis end.